

```

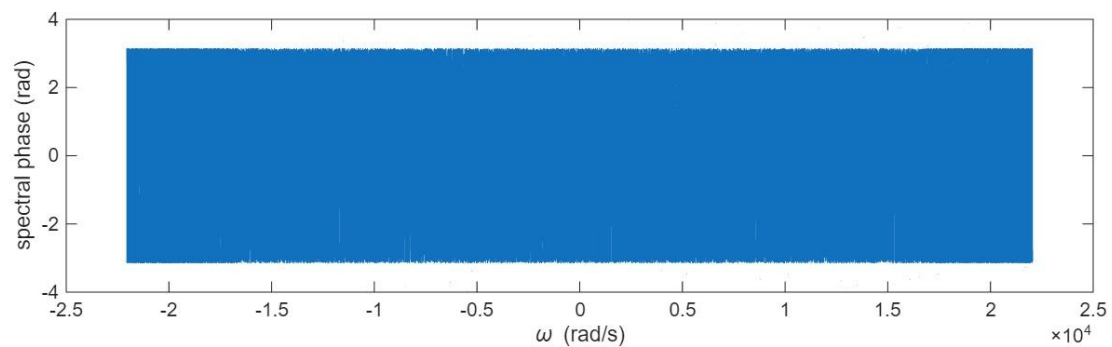
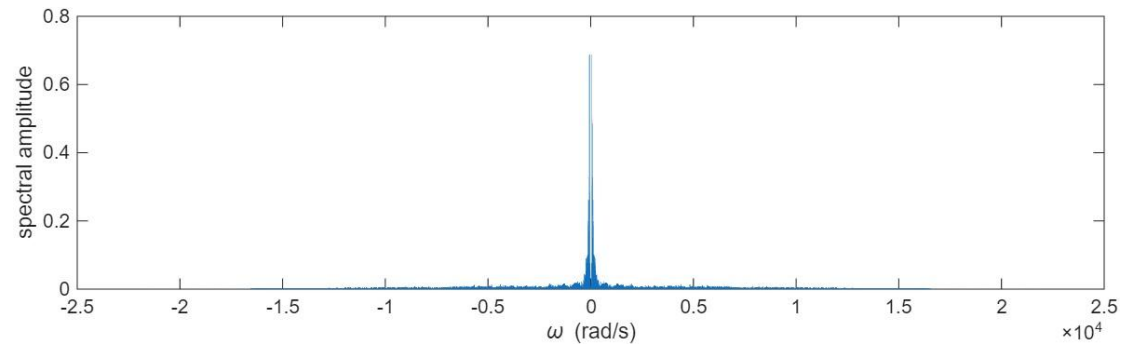
filename = '..\bad_apple!!.mp3';
[y, FS] = audioread(filename);
signal=y(FS*10+1:FS*20);
t = (0:length(signal)-1)/FS;

```

```

[omega,spectrum] = fft_PolyU(t,signal)

```



```

cutoff_freq = 500;
H = zeros(size(spectrum));
for lowpass_indices = find(omega >= -cutoff_freq & omega <= cutoff_freq)
    H(lowpass_indices) = 1;
end

```

```

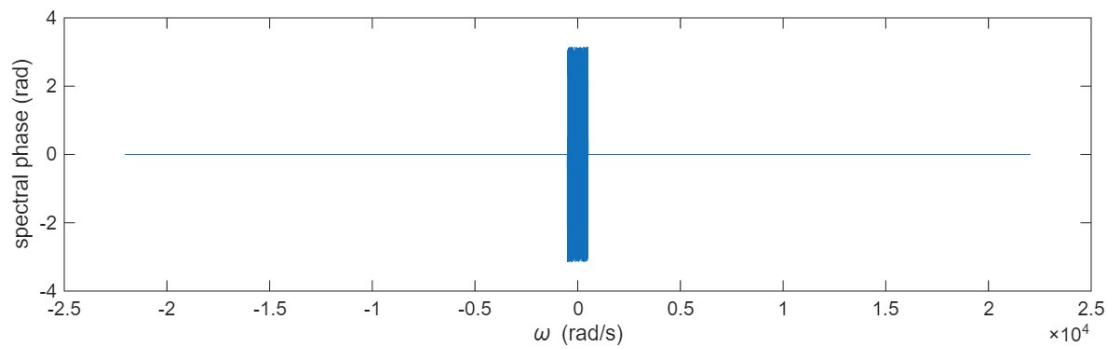
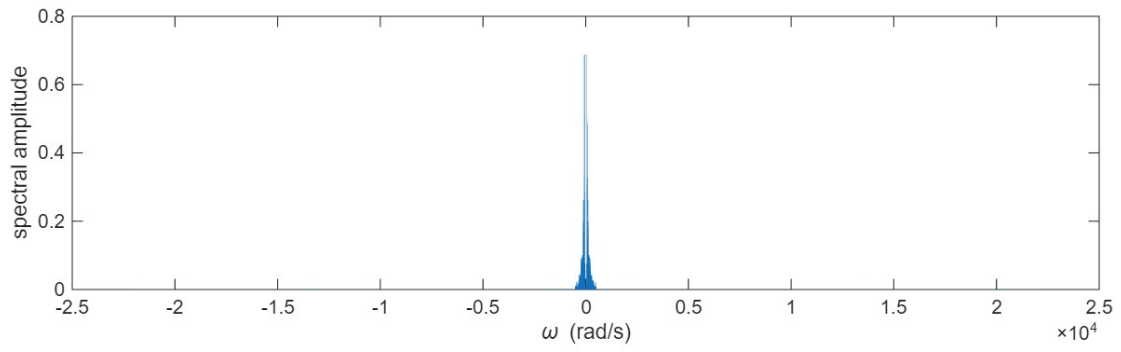
spectrum = spectrum .* H;
figure;
subplot(2,1,1);
plot(omega,abs(spectrum));
xlabel('\omega (rad/s)');
ylabel('spectral amplitude');

```

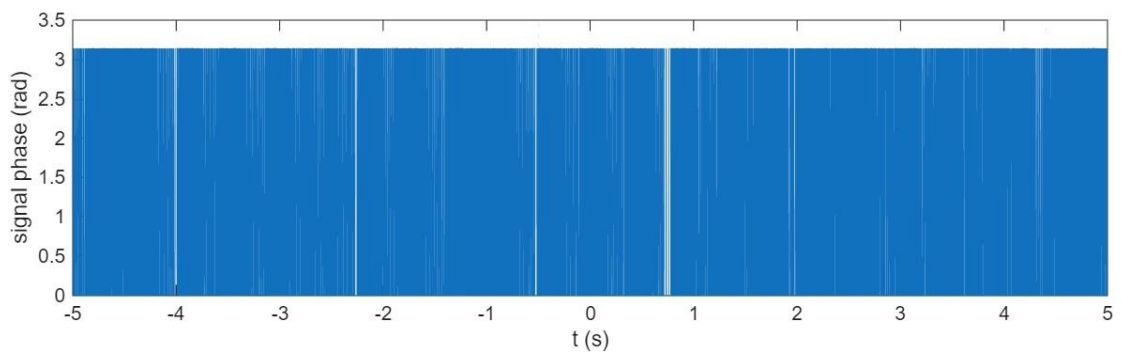
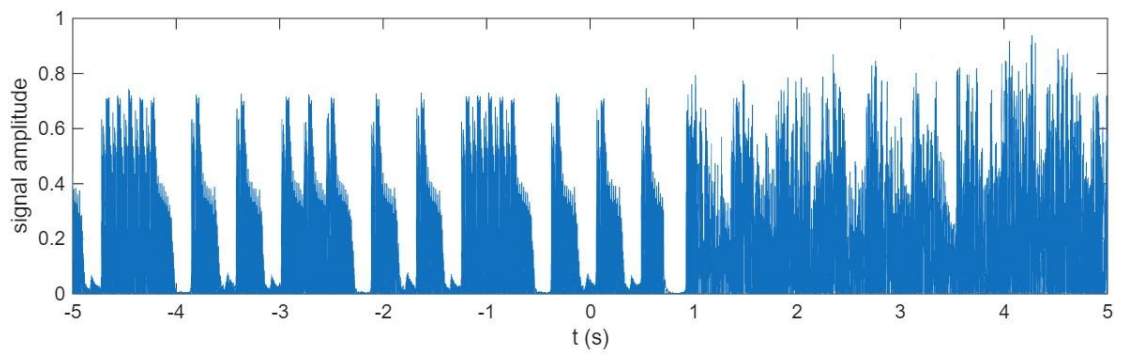
```

subplot(2,1,2);
plot(omega,angle(spectrum));
xlabel('\omega (rad/s)');
ylabel('spectral phase (rad)');

```



```
[t,signal] = ifft_PolyU(omega,spectrum)
```



```
player = audioplayer(signal,FS);  
play(player)
```

```
audiowrite('lowpass_500Hz.mp4',signal,FS)
```